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FINAL YEAR PROJECT REPORT
BACHELOR OF ENGINEERING

AFDM Receiver Design for Underwater Acoustic Communication: From Joint OMP-MP Baselines to Frozen Decision-Level Correction

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2025–2026

Coursework Declaration and Feedback Form

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Declaration of Originality and Submission Information	
I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements. *Signed (Student):*	UESTC4006P TEC-IT.COM
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Abstract

This final report summarizes a two-stage AFDM receiver project for underwater acoustic communication. The midterm stage established a MATLAB end-to-end AFDM-UWA simulation chain and a joint OMP-MP receiver that used one-shot OMP support locking followed by iterative message passing and least-squares gain refinement. For consistency, the final report does not mix those MATLAB figures with newer Python experiments; instead, all numerical claims are unified to the latest Python BER protocol. Under that protocol, the strongest traditional receiver from the joint OMP family is the strict traditional scaffold TRADPARAM FIXED, which improves the historical JOINTOMPMPbaseline at 15, 20, and 25 dB. The later project stage then revisited learned detector-side improvement. After correcting the evaluation pipeline, the original mean-residual distillation story collapsed, showing that the real recoverable headroom lies in sparse decision-boundary corrections rather than smooth residual regression. Guided by this diagnosis, the final method is a lightweight frozen direct-bit corrector that uses observable receiver trajectory features only. Trained once and frozen, it improves the mean BER from 6.3553×10^{-2} to 6.2493×10^{-2} on untouched nominal caches, remains positive under higher Doppler severity, integer-Doppler structural shift, and lower pilot power, and yields relative goodput gains up to 11.33% for 256-bit packets. A severe path-count shift remains an explicit failure boundary. The report therefore presents a coherent project trajectory: a strong traditional midterm baseline, a corrected negative result on residual distillation, and a final scope-bounded post-hoc correction method with reproducible gains.

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1 Introduction and Project Scope

Underwater acoustic (UWA) communication combines long delay spread, severe Doppler variation, and tight complexity constraints, which makes robust transceiver design difficult. AFDM is attractive in this setting because it offers a structured delay-Doppler representation that can support sparse channel reasoning while remaining compatible with modern model-driven receiver design ideas [1–3]. The challenge for this project was not only to build a strong AFDM receiver, but also to understand which part of the receiver stack is genuinely worth improving.

The project evolved in two phases. The midterm phase focused on a classical joint receiver implemented in MATLAB for an AFDM-UWA simulation chain. That stage produced a one-shot OMP support estimator followed by iterative MP-based detection and LS-based gain refinement. The final phase moved to the current Python codebase, where the same line of work was re-evaluated more carefully and later extended with learned post-hoc correction. This transition matters because the Python pipeline became the authoritative evaluation environment for the final report.

This final report therefore has two goals:

- record the strongest traditional result that grew out of the midterm `Joint OMP-MP` work, and
- explain how the project pivoted from a failed residual-distillation idea to a more reliable frozen decision-level corrector.

The project contributions are summarized as follows.

- A complete midterm AFDM-UWA receiver family was built around one-shot OMP support locking and iterative MP/LS refinement. In the final Python-aligned evaluation, the strongest traditional representative of that family is TRADPARAM FIXED, which consistently improves the older JOINTOMPMPbaseline.
- A corrected evaluation pass showed that the original detector-side mean-residual distillation hypothesis was not scientifically valid: once the evaluation bug was removed, the headline BER gain disappeared.
- A final train-once, frozen direct-bit corrector was introduced on top of the learned backbone CURRENTJOINT. It uses only observable receiver trajectory features, improves BER on untouched nominal caches, transfers across several AFDM regime shifts, and produces meaningful packet-level goodput gains.

The rest of the report follows the same project logic. section 2 identifies the correct midterm baseline and aligns it with the latest Python BER numbers. section 3 explains why the learned residual route was rejected and introduces the final frozen corrector. section 4 reports the main evaluation results. section 5 compares the main methods and clarifies their strengths and limitations. section 6 concludes with the final project outcome and next steps.

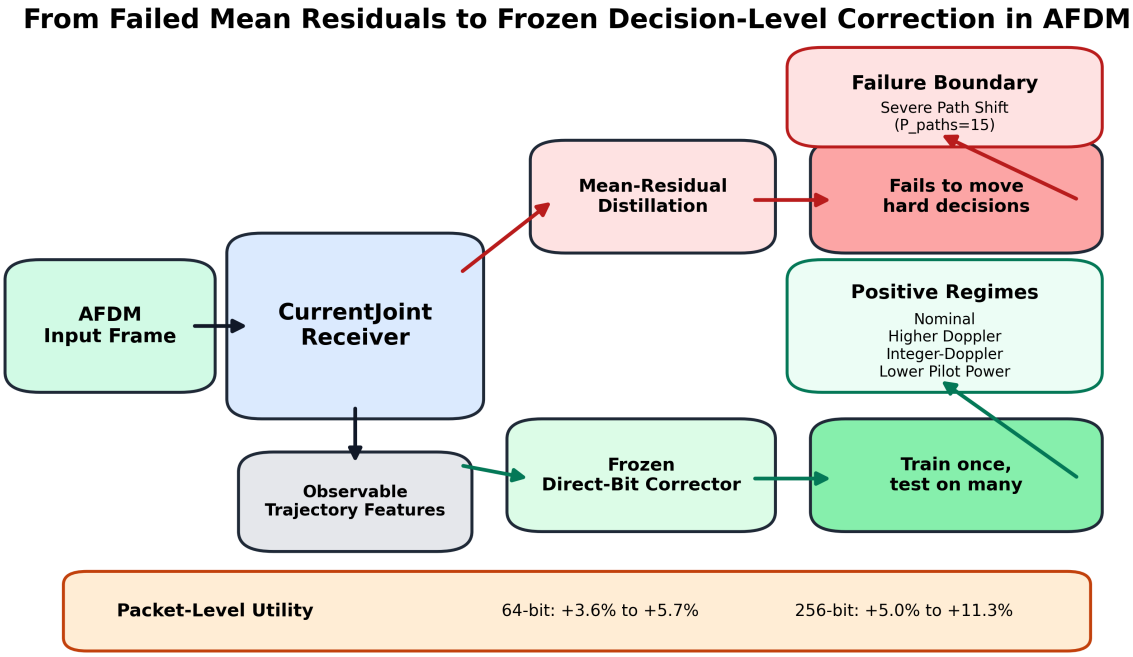


Figure 1: Project trajectory in the final report. The work starts from a joint OMP-MP traditional receiver, then pivots away from a failed mean-residual distillation idea, and ends with a frozen decision-level corrector that provides scope-bounded gains on top of the learned AFDM backbone.

Table 1: Python-aligned BER results for the joint OMP family at 15, 20, and 25 dB. These are the authoritative traditional-baseline numbers used in the final report.

Method	15 dB	20 dB	25 dB
JOINTOMPMP	1.8291×10^{-2}	1.0215×10^{-2}	5.4980×10^{-3}
TRADPARAM FIXED	1.6377×10^{-2}	8.8965×10^{-3}	5.0684×10^{-3}
TRADPARAM CLASSICAL_FULL	1.6709×10^{-2}	8.8086×10^{-3}	5.3516×10^{-3}

2 Midterm Traditional Receiver and Python-Aligned Baseline

2.1 What the midterm MATLAB work implemented

The midterm project artifacts, especially `AFDM_UWA_acoustic.m` and the interim report `2840210g_Guo_UESTC4006P_Interim_2026.pdf`, show a clear receiver design: the UWA AFDM chain used superimposed pilots, estimated a sparse delay-Doppler support by OMP, and then refined the receiver iteratively through MP detection plus LS gain updates on the fixed support. In other words, the midterm contribution was not a generic OMP detector, but a *joint OMP-MP* family of traditional iterative receivers.

That historical point matters because several later code paths exist in the repository. For the final report, the question is therefore not "did the project ever have a joint OMP baseline," but rather "which current baseline most faithfully represents that midterm work under the latest evaluation protocol." The answer is the strict traditional scaffold family in the Python codebase, not the later learned receiver variants.

2.2 Why this report uses Python BER numbers

The interim report used the MATLAB simulator and reported BER/NMSE trends from that environment. The current repository later moved to a Python pipeline with different logging, reproducibility controls, and corrected evaluation rules. Mixing MATLAB absolute BER values with final Python results would make the final report internally inconsistent. For that reason, this report keeps the MATLAB material as historical context only and unifies all numerical claims to the latest Python BER protocol.

2.3 Which traditional baseline should represent the midterm result

The final Python verification distinguishes three traditional receivers from the same family:

- JOINTOMPMP: the historical Python classical baseline,
- TRADPARAM FIXED: a strict traditional modernized scaffold with fixed hand-tuned parameters,
- TRADPARAM CLASSICAL_FULL: the same traditional scaffold with learnable classical scalars enabled.

Table 1 shows that TRADPARAM FIXED is the strongest traditional baseline to represent the midterm result in this final report. It beats JOINTOMPMPat all three verified SNR points and

reduces the mean BER from 1.1335×10^{-2} to 1.0114×10^{-2}, which is about a 10.8% relative reduction. By contrast, TRADPARAM CLASSICAL_FULL improves the 20 dB point slightly more than TRADPARAM FIXED, but it is worse at 15 and 25 dB, so it is not the cleanest or strongest summary of the traditional line.

The practical conclusion is straightforward: the midterm project result should be described as the *joint OMP-MP traditional receiver family*, and the final report should use TRADPARAM FIXED as its best current numerical representative. This choice preserves the spirit of the MATLAB midterm work while avoiding outdated or mismatched BER numbers.

2.4 Strengths and limitations of the traditional line

The traditional midterm receiver family has two clear strengths. First, it is highly interpretable: OMP finds a sparse support, MP handles symbol inference, and LS refines gains on a fixed low-dimensional parameter set. Second, it remains a strong low-SNR anchor; earlier freeze analyses in the project showed that the strict traditional scaffold is particularly stable in the lower-SNR regime.

Its limitation is equally clear. Even after the scaffold modernization that produced TRADPARAM FIXED, the traditional line still reaches a ceiling. Later learned backbones and post-hoc correction methods became attractive because they can exploit internal receiver-state information that the strict traditional scaffold does not explicitly model.

3 Diagnostic Pivot to the Final Frozen Corrector

3.1 Why the original learned route was rejected

After the traditional midterm line, the project moved to a stronger learned AFDM backbone, CURRENTJOINT, and asked whether a lightweight detector-side learning module could further improve BER. The first idea was mean-residual distillation: use teacher guidance to predict a smooth detector residual and apply it as a post-processing correction. This looked promising before the evaluation pipeline was cleaned up.

That story did not survive corrected evaluation. Once the same-test-set bug was removed, the base learned receiver, the variance-recalibrated variant VX_RECAL_V1, and the cleaned mean-residual model DISTILL_PURE_ITER0 became almost indistinguishable in BER. In other words, the apparent gain came from the evaluation pipeline rather than from a real receiver improvement.

The key diagnostic question then changed from "how do we regress the oracle residual" to "what kind of decision error is still recoverable." A decision audit on the corrected held-out cache gave three important numbers:

- base BER: 6.0352×10^{-2},
- teacher disagreement rate: 5.2051×10^{-2},
- oracle-beneficial flip rate: 3.1543×10^{-2}.

These values show that useful headroom still exists, but it is concentrated in a sparse subset of bits near the decision boundary. That structure is poorly matched to smooth residual regression.

3.2 Frozen direct-bit correction

The final method therefore changes the learning target. Instead of predicting a detector residual, it predicts whether the current hard decision should be corrected. The input is a 12-dimensional observable feature vector built from the initial and final receiver states:

$$\mathbf{z} = \left[\ell^{(T)}, |\ell^{(T)}|, \ell^{(0)}, |\ell^{(0)}|, \ell^{(T)} - \ell^{(0)}, |\ell^{(T)} - \ell^{(0)}|, v_x^{(T)}, v_x^{(0)}, v_x^{(T)} - v_x^{(0)}, \sigma^{(T)}, \sigma^{(0)}, \mathrm{SNR}\right]. \quad (1)$$

These features come entirely from the deployed receiver trajectory, so the corrector does not need teacher trajectories or split-specific adaptation at test time. The actual corrector is a small multilayer perceptron with two hidden layers of sizes 32 and 16. It is trained once on a dedicated cache and then frozen.

This "train once, test on many" rule is central to the final claim. It makes the model a genuine post-hoc correction module rather than an evaluation-time probe. Before committing to this final design, the project also tested a split-specific observable direct-bit probe, which showed gains between 6.8359×10^{-4} and 9.2773×10^{-4} on fresh caches. That probe established that the direction was viable; the frozen corrector then turned it into a reportable final method.

3.3 What changed scientifically

The scientific lesson of this pivot is stronger than a simple method substitution. The corrected project conclusion is that oracle headroom in this AFDM receiver does exist, but it is not well described by a mean residual. The recoverable part is sparse, local, and decision-level. That is

why the final method is intentionally modest: it leaves the backbone CURRENTJOINTunchanged and adds only a frozen correction head whose role is easy to interpret.

Table 2: Chronological view of the main methods in the project. The table mixes different evaluation protocols on purpose: it records project evolution and scientific role, not a single apples-to-apples BER leaderboard.

Method	Family	Evaluation role	Headline outcome
Joint OMP-MP (MATLAB interim)	Midterm traditional prototype	historical origin of the joint receiver line	one-shot OMP support locking plus MP/LS refinement in `AFDM_UWA_acoustic.m`
JOINTOMPMP	Historical Python classical baseline	anchor inside the final Python codebase	retained as the reference classical iterative baseline
TRADPARAM FIXED	Strict traditional modernized scaffold	final-report representative of the midterm line	improves JOINTOMPMPat 15, 20, and 25 dB
CURRENTJOINT	Learned backbone	deployment backbone for the final stage	base receiver for all final correction studies
VX_RECAL_V1	Detector calibration control	corrected negative control	no meaningful BER change after correction
DISTILL_PURE _ITER0	Mean-residual distillation	corrected negative result	headline gain collapses once evaluation is fixed
FROZENBIT CORRECTOR	Final method	train-once, frozen post-hoc correction	nominal BER gain 1.0590×10^{-3} and positive transfer across three shifts

4 Experimental Evaluation

4.1 Method evolution

Table 2 shows how the project matured. The midterm line produced a strong and interpretable traditional receiver. The final line produced a narrower but more credible learned correction story. The most important methodological change is that the project stopped treating all later routes as equally plausible: calibration-only fixes and mean-residual distillation are now explicitly recorded as negative controls rather than presented as candidate final methods.

4.2 Main frozen-corrector results

The final method is evaluated on top of CURRENTJOINTusing a corrected train-once/test-on-many protocol. On three untouched nominal caches, the mean BER improves from 6.3553×10^{-2} to 6.2493×10^{-2}, which gives an average gain of 1.0590×10^{-3}. The gain standard deviation across the three caches is only 5.5328×10^{-5}, which supports the claim that the result is reproducible rather than a single-split accident.

Figure 2 summarizes the transfer results. The same frozen checkpoint remains positive under higher Doppler severity, an integer-Doppler structural shift, and lower pilot power:

- higher Doppler severity: 8.4310×10^{-4} mean BER gain,
- integer-Doppler structural shift: 1.3249×10^{-3} mean BER gain, with 95% CI $[1.1621 \times 10^{-3}, 1.4958 \times 10^{-3}]$,

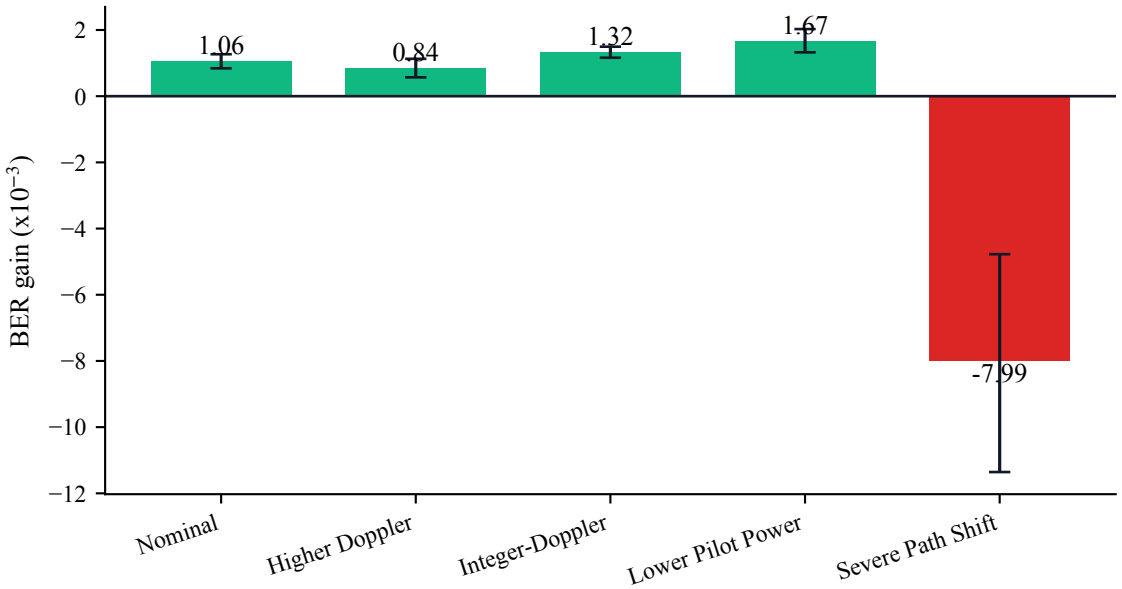


Figure 2: BER gains of the frozen direct-bit corrector across evaluation regimes. The positive regimes remain above zero, while the severe path-count shift becomes a clear negative boundary.

- lower pilot power ($\rho = 0.4$): 1.6699×10^{-3} mean BER gain.

The same figure also shows the explicit failure boundary: under a severe path-count shift with $P_{\text{paths}} = 15$, the mean BER change becomes -7.9883×10^{-3}. This negative regime is kept intentionally because it defines the scope of the method honestly.

4.3 Why the final method is more than LLR thresholding

Figure 3 answers an important reviewer question: is the final method just a confidence threshold on the last LLR? The answer is no. The full feature set reaches a mean gain of 1.1133×10^{-3}, whereas the LLR-only variant falls to 2.1267×10^{-4}. Removing final LLR, v_x, effective noise, or SNR each causes a measurable drop. The corrector therefore benefits from joint evidence across trajectory evolution and uncertainty, not just one confidence scalar.

4.4 Packet-level practical significance

One remaining concern after the BER study was practical significance. The report therefore converts the bit-level gains into contiguous-packet success and a simple ARQ-style goodput proxy. For 64-bit packets, the relative goodput gain ranges from 3.62% to 5.72% across the positive regimes. For 256-bit packets, the gain ranges from 4.97% to 11.33%. Under the severe path-count shift, the packet-level benefit disappears, which is consistent with the BER regression.

Taken together, the experiments support a final claim that is smaller but stronger than the early project hypothesis: the project did not discover a universally better learned AFDM detector, but it did identify a reliable, frozen, decision-level post-hoc correction method for the current AFDM receiver family.

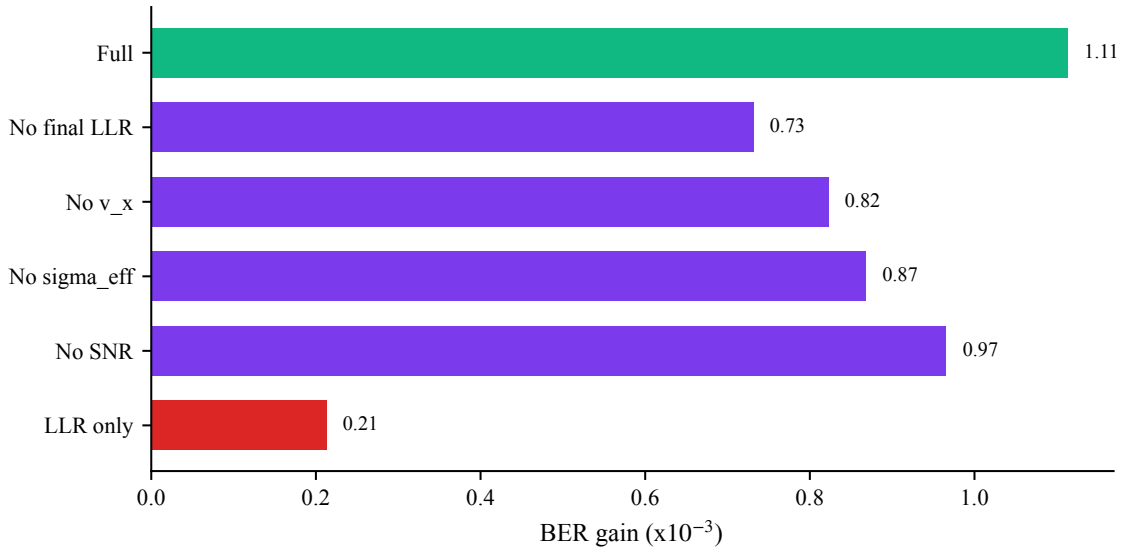


Figure 3: Feature-group ablation for the frozen corrector. The full feature set is strongest, while an LLR-only version keeps only a small part of the gain.

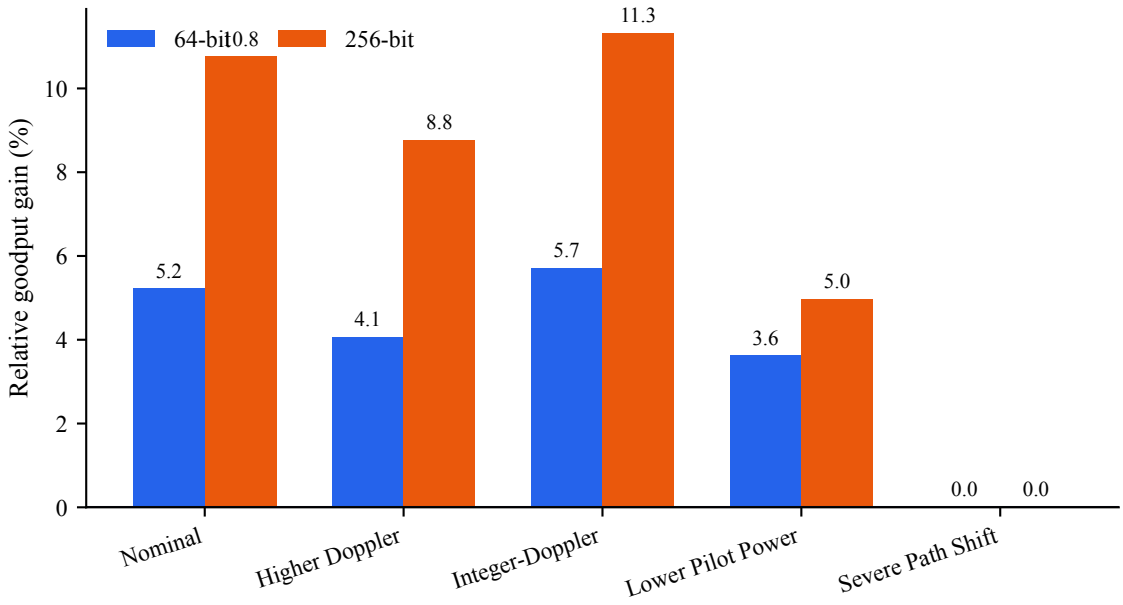


Figure 4: Relative goodput gains for 64-bit and 256-bit contiguous packets. Small BER gains become more visible at the packet level in the positive regimes.

5 Discussion: Which Methods Worked Best and Why

The project now has two different "best" methods, depending on what question is being asked.

If the question is *which traditional method best represents the midterm result*, the answer is TRADPARAM FIXED. It is the strongest strict traditional baseline in the current Python verification and is the most faithful final-report replacement for the original MATLAB joint OMP-MP receiver. Its main advantages are interpretability, low-risk behavior, and strong low-SNR stability. Its main limitation is that it still inherits the ceiling of the traditional scaffold.

If the question is *which final method gives the best overall result in the current final-stage study*, the answer is CURRENTJOINT plus the frozen direct-bit corrector. This is the method that survives corrected evaluation, improves untouched nominal BER, transfers across three regime shifts, and yields packet-level goodput gains. Its advantage is that it achieves extra performance without redesigning the entire receiver. Its limitations are equally important: the absolute BER gain is modest, all positive evidence still lies within the current AFDM simulator family, and the method fails under a severe path-count shift.

The project also contains two routes that are useful mainly because they failed. The calibration-only route VX_RECAL_V1did not change BER meaningfully after the evaluation was corrected. The mean-residual distillation route DISTILL_PURE_ITER0 is even more important as a negative result, because it taught the project that the remaining headroom is sparse and decision-level rather than smooth and regression-friendly. In hindsight, this negative result was one of the most valuable turning points in the work.

From a project-report perspective, this leads to a clean comparison of strengths and weaknesses:

- JOINTOMPMP: simplest classical anchor and easy to explain, but clearly weaker than the modernized strict traditional scaffold.
- TRADPARAM FIXED: best traditional baseline and the right summary of the midterm achievement, but still bounded by a traditional ceiling.
- CURRENTJOINT: strongest backbone for the final stage, but by itself it does not recover the remaining decision-level headroom.
- FROZENBITCORRECTOR: best final method in the corrected study, because it adds reproducible gains without retraining on every test cache; however, it remains scope-bounded rather than universal.

This interpretation also explains why the final report is no longer framed as "a new AFDM receiver architecture." The stronger story is a project trajectory: build a strong traditional baseline, test a plausible learned idea, reject it when the evidence collapses, and keep only the post-hoc correction mechanism that survives corrected evaluation.

6 Conclusions and Future Work

This final report consolidates the project into a single consistent narrative. The midterm stage produced a genuine technical result: a joint OMP-MP traditional AFDM receiver for UWA communication. In the final Python-aligned evaluation, the best representative of that line is TRADPARAM FIXED, which consistently improves the older JOINTOMPMPbaseline and should therefore be treated as the authoritative traditional result in the final report.

The later stage of the project then asked a harder question: can detector-side learning improve an already strong AFDM receiver? The corrected answer is mixed but scientifically useful. Mean-residual distillation does not hold up under corrected evaluation. However, once the problem is reframed as a sparse decision-level correction task, a small frozen bit corrector does deliver reproducible gains on untouched caches and several shifted regimes, with packet-level improvements that are easier to appreciate than the raw BER deltas.

The final outcome is therefore stronger than a simple list of positive numbers. The project now contains:

- a clear and reproducible traditional midterm baseline,
- an honest corrected negative result on residual distillation,
- a final scope-bounded frozen correction method that is simple, deployable, and experimentally supported.

The most valuable future work directions are also clear. First, the frozen corrector should be tested beyond the current AFDM simulator family. Second, the packet-level analysis should be extended to coded links and more realistic end-to-end system metrics. Third, the sharp failure under the severe path-count shift suggests that future correction modules may need a better way to recognize when they are out of regime.

References

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A Alignment of Interim MATLAB Work and Final Python Reporting

This appendix records how the midterm materials were mapped into the final report.

- The interim MATLAB file `AFDM_UWA_acoustic.m` implements the joint OMP-MP receiver family: one-shot OMP support locking, followed by MP detection and LS gain refinement on the fixed support.
- The interim PDF is used as historical evidence for project trajectory and method intent, but not as the source of final absolute BER values.
- All numerical claims in the final report are unified to the latest Python evaluation pipeline. Under that pipeline, TRADPARAM FIXED is the strongest strict traditional representative of the midterm receiver family and is therefore the traditional baseline used in the body of the report.
- The learned final-stage results use a different evaluation protocol on top of CURRENTJOINT. They should be interpreted as the final project stage, not as direct numerical replacements for the traditional 15/20/25 dB table.

This alignment rule avoids a common reporting error: mixing attractive historical MATLAB figures with later Python results that use different simulation assumptions, logging rules, or evaluation hygiene. The final report keeps the history, but only the corrected Python numbers are treated as authoritative.